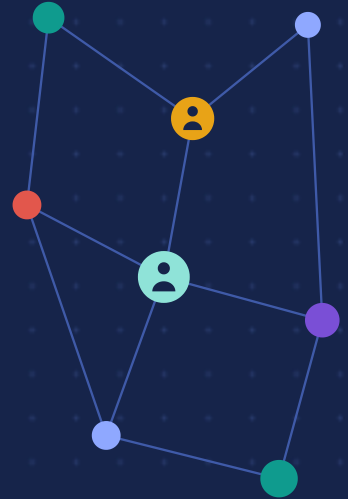


AP COMPUTER SCIENCE PRINCIPLES

WEEK 1

Collaboration & Computing Innovations

How teams build better technology



Name: _____

Class/Period: _____

Date: _____

1 • Learn

Read the short explanations and vocabulary.

2 • Try

Quick checks right after each new idea.

3 • Practice

Sort, match, and label real examples.

4 • Apply

Plan your own team's app like a developer.

p.2 Why Don't Programmers Just Work Alone?

p.3 The Words You Need

p.4 What Counts as a Computing Innovation?

p.5 Different Perspectives = Better Ideas

p.6 Don't Guess What Users Want

p.7 Pair Programming: Driver + Navigator

p.8 Working Together Online

p.9 Interpersonal Skills

p.10 Name That Skill

p.11 Apply It: Plan a Team App

p.12 Wrap-Up & Self-Check

Why Don't Programmers Just Work Alone?

Almost every app, game, and website you use was built by a team. Here's why.

SCENARIO

Your school wants a new app called **LunchLine** that shows the cafeteria menu and lets students pre-order lunch. Five students volunteer to build it. Each one notices something the others don't.

CODER

Priya

Has taken coding classes before.

"I can build the ordering system."

DESIGNER

Marcus

Loves art and graphic design.

"Buttons should be big and easy to tap."

INSIDER

Elena

Helps in the cafeteria after school.

"Orders must close by 10:00 a.m."

ALLERGY EXPERT

Dev

Has a peanut allergy.

"We need allergen labels on every item."

TRANSLATOR

Ana

Speaks Spanish at home.

"Some families will need a Spanish option."

If Priya built LunchLine **alone**, the code might work, but the app could be hard to use, miss the ordering deadline, leave out allergy info, and only work in English. Each teammate's **skills**, **experiences**, and **perspective** made the app better.

Different Skills

+

Different Experiences

+

Different Perspectives

↓

More Ideas

↓

Fewer Missed Problems

↓

Better Computing Innovation

THE BIG IDEA

Different perspectives help teams create **better computing innovations**.

When people with different talents and backgrounds **collaborate**, the team thinks of more ideas and catches problems that one person, or a team of people who are all alike, would miss.

The result is a product that works well for **more people**.

TRY IT

Why might a team with different skills create a better app than a team where everyone has exactly the same skills? Use an example from LunchLine if it helps.

The Words You Need

You don't need to memorize these word for word. Understand them well enough to explain each one in your own words.

■ Teamwork ■ Innovations & bias ■ Working with users ■ People skills

Collaboration

Working together with others to create something or solve a problem.

Ex: Four students split up the work of building a game.

Computing Innovation

An innovation that includes a **program** as an integral (essential) part of how it works.

Ex: A smartphone, a photo-editing app, or e-commerce.

Bias

A tendency or systematic error that can cause a computing system or decision to unfairly favor one group or outcome.

Ex: A voice app that understands some accents much better than others.

Diversity of Talents and Perspectives

The range of skills, backgrounds, experiences, and viewpoints that team members bring.

Ex: A team with a coder, an artist, and someone who knows the users well.

Consultation

Asking users or other stakeholders for their input, especially while planning and building.

Ex: Interviewing teachers before designing a grading app.

Feedback

Information or comments that help evaluate and improve an idea, product, or piece of code.

Ex: A tester says, "The Save button is hard to find."

Communication

Sharing information, ideas, needs, and problems clearly and often.

Ex: Posting a quick team update: "Login page is done!"

Pair Programming

Two programmers share one computer. One writes code; the other reviews it and thinks ahead. They switch roles.

Roles: Driver (types) + Navigator (guides).

Interpersonal Skills

The "people skills" that help a team work well together.

Includes: communication, consensus building, conflict resolution, negotiation.

Consensus Building

Helping the whole team reach a decision that everyone can agree to support, after every voice is heard.

Ex: Everyone shares an idea, then the team picks one together.

Conflict Resolution

Handling a disagreement or problem between people in a calm, respectful, productive way.

Ex: Two teammates talk it out after a misunderstanding.

Negotiation

Discussing different wants to find a compromise, where each side gives a little.

Ex: "You pick the colors, I'll pick the font."

TRY IT

Put a ★ next to the two words that are **newest** to you. Put a ✓ next to one word you have seen before in a club, sport, or job.

What Counts as a Computing Innovation?

DEFINITION

A **computing innovation** includes a **program** as an integral part of how it works.

"**Integral**" means *essential*. Ask yourself: **If you took away the program, would it still do its main job?** If the answer is **no**, it's a computing innovation. Computing innovations come in three types:



PHYSICAL

A device you can touch, run by a program.

- › Smartphone
- › Robot
- › Smart device (smart speaker, smartwatch)
- › Self-driving vehicle



SOFTWARE

Nonphysical: a program that runs on a device.

- › App
- › Video game
- › Photo editor
- › Word processor



NONPHYSICAL CONCEPT

A big idea or system that depends on computing.

- › E-commerce (online shopping)
- › Social networking

Not everything is a computing innovation. A pencil, a paper notebook, and a wooden chair are useful inventions, but no program is needed to make them work.

PRACTICE Spot the Innovation

Circle each item that is a computing innovation. Put an **X** through items that are not.

Smartphone

Pencil

Video game

Robot

Paper notebook

E-commerce

Photo editing app

Wooden chair

Label it: Write the **type** (Physical, Software, or Nonphysical Concept) for three of the items you circled.

Item: _____

Item: _____

Item: _____

Type: _____

Type: _____

Type: _____

Explain: Pick **one** item you did **not** circle. Why isn't it a computing innovation?

Different Perspectives = Better Ideas

Everyone sees the world from where they stand. Teams need more than one view.

Why diversity matters in computing

People bring different **backgrounds, experiences, skills, knowledge,** and **perspectives.** When a team includes many of these, it can:

- › come up with **more ideas**
- › notice **problems** others missed
- › spot and reduce **bias**
- › make a product useful for **more people**

What is bias?

Bias is a tendency or systematic error that makes a system unfairly favor one group or outcome.

In computing, bias is often **not on purpose.** It happens when the team only thinks about people *like themselves.*

Example: A fitness app is tested only by teammates who walk. It counts steps well, but it doesn't work for people who use wheelchairs.

SCENARIO · THE SCHEDULE APP

A student team builds **BellBoard**, an app that shows your class schedule and sends a reminder before each class.

Everyone on the team has the **exact same schedule:** 1st lunch, no zero period, every class on campus. The app works perfectly for them.

After the app is released, other students report problems:

- › Students with **2nd lunch** get reminders at the wrong times.
- › **Zero-period** classes don't appear at all.
- › Students who take a class at the **community college** can't enter it.
- › Reminders are **sound-only**, so a Deaf student misses them.

BellBoard

P1 Biology 8:00

P2 English 8:55

LUNCH (1st) 9:50

2nd lunch? ???

1. **What did the team miss?** Name at least one group of students the app didn't work well for.

2. **How could different perspectives have helped?** Who could the team have added or talked to?

3. **How does this connect to bias?** *Hint: Who does the app favor? Did the team mean to do that?*

Don't Guess What Users Want

Developers should talk with users and **potential** users (people who *might* use it), not just guess.

Consultation

Getting input from users and stakeholders. *"What do you need this app to do?"*

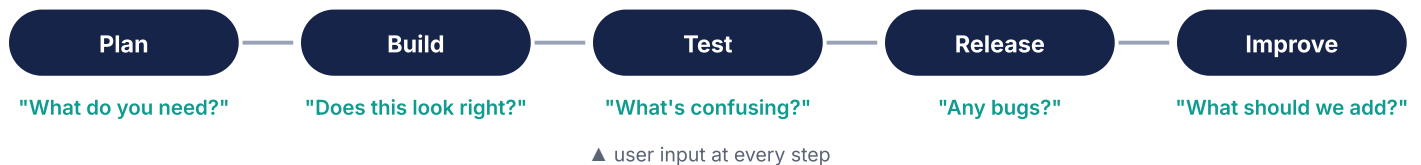
Feedback

Information that helps evaluate or improve something. *"This menu is confusing."*

Communication

Sharing information, ideas, needs, and problems back and forth. *"Here's what we changed."*

USERS CAN HELP AT EVERY STAGE, NOT JUST AT THE END



PRACTICE Match the Term

Write **CON** (Consultation), **FB** (Feedback), or **COM** (Communication) for each situation.

#	Situation	Term
1	Before writing any code, a developer interviews five teachers about what a homework app should do.	_____
2	A tester says, "The font is too small to read on my phone."	_____
3	The team posts an update telling users that a new dark mode is coming next week.	_____
4	A game company surveys players to ask which new levels they would want before designing them.	_____
5	A user rates the app 3 stars and writes, "It crashes when I upload photos."	_____

TRY Which Feedback Should the Developer Use?

Jordan is building **StudyBuddy**, an app that reminds students about homework. Early testers said:

- ☐ A. "I can't tell which assignments are due *today*." ☐ B. "Make the app purple. Purple is the best color."
- ☐ C. "The reminders don't work when my phone is on silent." ☐ D. "My parent wants to see my list too. Can it be shared?"

Check the **two** pieces of feedback that would help the most. Explain one choice:

Pair Programming: Driver + Navigator

A common way programmers collaborate, and one you'll use in this class.

2 + 1

Two programmers share one computer. One person writes the code while the other reviews, checks for problems, and thinks ahead. Partners **switch roles** often.



DRIVER

HANDS ON THE KEYBOARD

- › Types the code
- › Focuses on the **current line**
- › Explains what they're doing out loud
- › Listens to the navigator



NAVIGATOR

EYES ON THE BIG PICTURE

- › Reviews each line as it's typed
- › Catches typos and bugs
- › **Thinks ahead** to the next step
- › Asks questions and suggests ideas

Memory trick: Think of a road trip. The **driver** steers the car. The **navigator** reads the map and says, "Our turn is coming up!" You need both to get there safely.

Why it works

- › Two sets of eyes → **fewer bugs**
- › Instant **feedback** while coding
- › Partners **learn from each other**

Good partner habits

- › Switch roles every 10–15 minutes
- › Be kind and specific: "Line 4 needs a closing bracket."
- › Navigators: **don't grab the keyboard!**

PRACTICE Driver or Navigator?

Write **D** for Driver or **N** for Navigator.

#	Action	D or N
1	Types <code>score = score + 1</code> into the editor.	_____
2	Says, "Wait, I think we forgot to reset the score when the game restarts."	_____
3	Suggests, "After this, we should add the timer."	_____
4	Explains out loud, "I'm adding an if-statement here to check for a win."	_____

Why do you think partners should switch roles?

Working Together Online

Teammates don't have to sit in the same room, or even the same country. Online tools let them work together from anywhere.

Online tools support collaboration by letting programmers **share** ideas and documents and **give feedback** on each other's work. *You do **not** need to memorize any app names. Focus on what the tools **do**.*

Share Work

Put files and code in one shared place so everyone has the latest version.

Communicate

Chat, message, or video call to ask questions and share updates.

Give Feedback

Leave comments and suggestions directly on someone's work.

Edit Together

Several people edit the same document or code file at the same time.

Track Tasks

Use a task board to see who is doing what and when it's due.

Build Together

Shared coding platforms save every change so a team can combine work and undo mistakes.

TRY Which Tool Feature Helps?

Choose the feature from the boxes above that best solves each problem.

#	Team problem	Tool feature
1	Two teammates keep doing the same task by accident, and nobody finishes the menu screen.	_____
2	A teammate is home sick but wants to suggest changes to the project plan.	_____
3	The team has three different copies of the code and doesn't know which one is newest.	_____
4	Someone accidentally deletes a working feature and needs to get the old version back.	_____
5	Three students need to write the project's user guide together tonight from their own homes.	_____

One benefit of working online with a team:

One challenge of working online, and how to handle it:

Interpersonal Skills: How Good Teams Work

Great code isn't enough. Effective teams practice **interpersonal skills**, the "people skills" of collaboration.

1

Communication

Sharing information, ideas, needs, and problems **clearly, kindly, and often**.

Example: "I'm stuck on the login page. Can someone look at it with me?"

2

Consensus Building

Making sure **every** team member's ideas are heard so the whole group can agree on a decision.

Example: Each person pitches an app name, then the team discusses and agrees on one everyone can support.

3

Conflict Resolution

Handling a **disagreement or tension** between people calmly and respectfully so the team can move forward.

Example: Two teammates are upset that the other "took over." They talk it out and agree to take turns leading.

4

Negotiation

Discussing different **wants** to reach a **compromise**, a middle ground where each side gives a little.

Example: "We'll add your leaderboard idea now, and my chat feature in version 2."

How are they different?

They can overlap, but each skill answers a **different question**:

Skill	The question it answers	Key clue words
Communication	"Does everyone know what's going on?"	updates, sharing, asking, explaining
Consensus Building	"Does the whole group agree on this decision?"	everyone's input, vote, group decision
Conflict Resolution	"How do we fix a problem between people ?"	upset, argument, tension, misunderstanding
Negotiation	"What trade or middle ground works for both sides?"	compromise, trade, "if you... then I..."

Tricky pair: Conflict Resolution vs. Negotiation

Conflict resolution is about fixing a *problem between people* (hurt feelings, arguments). **Negotiation** is about finding a *deal* when people want different things. Sometimes negotiation is one *tool* used to resolve a conflict.

TRY IT

In your own words: What's the difference between **consensus building** and **negotiation**?

Name That Skill

Read each team moment. Which interpersonal skill is being used **most**? Use the clue words from page 9.

COM
Communication

CB
Consensus Building

CR
Conflict Resolution

NEG
Negotiation

#	Team moment	Skill
1	Kai sends the group chat a message: "I finished the sound effects and uploaded them to our shared folder."	_____
2	Lena wants the game to have 10 levels. Omar wants 3. They agree on 5 levels for now and more later.	_____
3	Two teammates argued loudly yesterday. Today they meet, each explains how they felt, and they agree on how to share work fairly.	_____
4	The team leader notices Mia hasn't spoken, so she asks, "Mia, what do you think?" before the group makes a final choice.	_____
5	Sam tells the team, "I won't be able to finish my part by Friday. I'm going to need help."	_____
6	"If you write the instructions page, I'll fix the bug in your character movement code."	_____
7	The group lists all five logo ideas, discusses each one, and keeps talking until everyone is okay with the final design.	_____
8	Jay is frustrated because his idea was laughed at. The team apologizes and agrees on a rule: "No putting down ideas."	_____

APPLY What Would You Say?

Your team is building a quiz app. **Riley** keeps changing everyone else's code without asking, and **Tasha** is getting upset. The team is falling behind.

- a. Which interpersonal skill does this team need most right now? _____
- b. Write **exactly what you would say** to the group to help:

Apply It: Plan a Team App

Now you think like a developer. Use at least **three** vocabulary words from page 3 in your answers.

YOUR MISSION

Your team of four is designing **FirstWeek**, an app to help **new 9th graders** find their way around your school during their first week.

1. Build your team. Describe four teammates with **different** strengths or perspectives. What does each one add?

Teammate strength or background	What they add to FirstWeek
<i>e.g., a student who uses a wheelchair</i>	<i>e.g., makes sure maps show elevators and ramps</i>

2. Consultation. Name **two** groups of people you should talk to *before* building, and one question you'd ask them.

3. Watch for bias. Describe one way FirstWeek could accidentally work better for some students than others, and how your team would prevent it.

4. Is FirstWeek a computing innovation? Circle one: **YES** / **NO**. Type: _____

5. Teamwork plan. How will your team use **pair programming** and **online tools** to build the app?

6. Feedback. After a test run, a 9th grader says, "I couldn't find the cafeteria on the map." What should your team do next?

Wrap-Up & Self-Check

Look back at what you learned this week.

1 Big Ideas Summary

Collaboration brings together different talents and perspectives to build better computing innovations.

Diverse teams help **avoid bias** and create products that work for more people.

A **computing innovation** includes a program as an integral part of its function. It can be physical, software, or a nonphysical concept.

Developers should use **consultation, feedback, and communication** with users throughout development.

Pair programming and **online tools** are common ways programmers collaborate.

Effective teams practice **communication, consensus building, conflict resolution, and negotiation**.

2 Exit Ticket

a. Finish the sentence: *"Teams with different perspectives make better technology because..."*

b. Describe a real app or game you use. What is **one piece of feedback** you would give its developers?

3 How confident are you?

Circle 1 (not yet) to 4 (I could teach it).

I can explain what a **computing innovation** is and give examples of all three types.

1 2 3 4

I can explain how **diverse perspectives** improve an innovation and help avoid **bias**.

1 2 3 4

I can explain why developers **consult users** and use **feedback**.

1 2 3 4

I can describe the **driver** and **navigator** roles in pair programming.

1 2 3 4

I can tell the difference between the **four interpersonal skills**.

1 2 3 4

AP CSP COURSE FRAMEWORK ALIGNMENT · BIG IDEA 1: CREATIVE DEVELOPMENT (CRD)

CRD-1.A Explain how computing innovations are improved through collaboration (pp. 2, 4–6, 11). **CRD-1.B** Explain how computing innovations are developed by groups of people (pp. 7–8). **CRD-1.C** Demonstrate effective interpersonal skills during collaboration (pp. 9–10).